

Narrative Nurse/Clinician Report — Vital Trends & Episode Review

Source: hourly HR/RR min/max/avg | Period: 2024-05-01 → 2024-06-30

Purpose: Provide shift- and clinician-friendly situational awareness by highlighting clusters of abnormal heart rate and respiratory rate that can precede clinical deterioration. This report is decision-support only and must be interpreted with symptoms, medications, SpO₂, blood pressure, and clinical exam.

High-alert context (user-provided): Patient experienced serious cardiac distress around **June 24, 2024** and died on **July 1, 2024**. The goal here is to make potential precursors visually obvious and operationally actionable.

Why these episodes matter: Sustained **bradycardia** (especially HR minima <45 bpm) can reflect conduction disease, medication effect, or sleep-related physiology—but when it is new, prolonged, or paired with respiratory stress it may represent worsening hemodynamics. **Tachycardia** and/or **tachypnea** clusters can signal infection, pain, hypoxia, volume overload/CHF, pulmonary pathology, or atrial arrhythmia. Patterns that intensify over days (baseline drift upward in RR or downward in HR) should prompt earlier evaluation.

Recommended bedside/RPM response when flagged clusters appear:

- 1) **Validate** (rule out artifact): confirm pulse/respirations; check sensor placement & motion; note low-sample hours.
- 2) **Assess:** BP, SpO₂, work of breathing, chest pain, dizziness/syncope, mental status, edema/weight change, fever.
- 3) **Reconcile medications:** AV node blockers (β-blockers, non-DHP calcium channel blockers), antiarrhythmics, sedatives/opioids.
- 4) **Escalate:** sustained HR <45, new/worsening brady clusters, tachycardia with dyspnea/hypotension, tachypnea with hypoxia → notify MD/cardiology/respiratory per facility protocol.

Signal summary (hours)	Count
Bradycardia (HR avg <50)	20
Severe bradycardia (HR min <45)	26
Tachycardia (HR avg >100)	7
Tachypnea (RR avg >24)	14
Low-confidence (cnt <30)	29

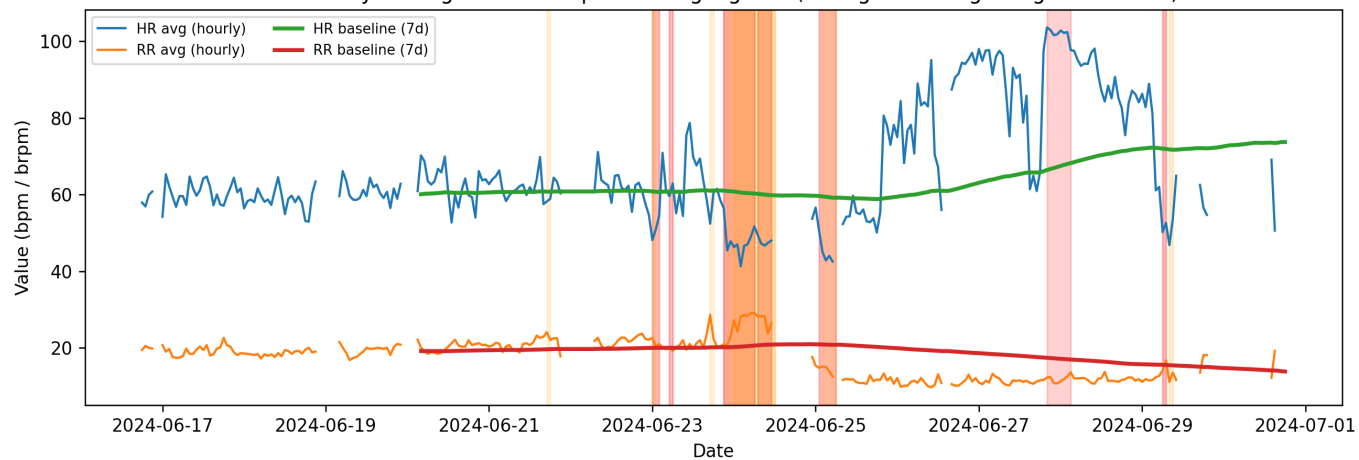
High-priority review near June 24 (top detected clusters)

Condition	Window	Hrs	Vitals snapshot	Who to notify
Severe bradycardia	2024-06-23 21:00 → 2024-06-24 05:00	9	Mean HR 47 Min HR 40 Mean RR 26 Max HR 100	MD/CC Cardiology/Respiratory if symptomatic
Severe bradycardia	2024-06-24 07:00 → 2024-06-24 10:00	4	Mean HR 48 Min HR 42 Mean RR 27 Max HR 98	MD/CC Cardiology/Respiratory if symptomatic
Tachypnea	2024-06-24 00:00 → 2024-06-24 09:00	10	Mean HR 47 Min HR 40 Mean RR 28 Max HR 100	MD/CC Cardiology/Respiratory if symptomatic
Bradycardia	2024-06-23 22:00 → 2024-06-24 05:00	8	Mean HR 46 Min HR 40 Mean RR 26 Max HR 100	MD/CC Cardiology/Respiratory if symptomatic
Bradycardia	2024-06-24 07:00 → 2024-06-24 11:00	5	Mean HR 48 Min HR 42 Mean RR 27 Max HR 98	MD/CC Cardiology/Respiratory if symptomatic
Severe bradycardia	2024-06-25 01:00 → 2024-06-25 05:00	5	Mean HR 45 Min HR 40 Mean RR 14 Max HR 99	MD/CC Cardiology/Respiratory if symptomatic

14-day episode plot (more readable; higher concern shaded in red/orange)

Darker shading corresponds to higher concern (severity score incorporates duration, severity, and co-occurrence).

Last 14 Days — High-Concern Episodes Highlighted (stronger shading = higher concern)



Appendix — Episodic Event Table

Rows with **severity score** ≥ 9 are shaded red (higher concern).

Condition	Start	End	Hrs	Sev	MeanHR	MinHR	MeanRR	MaxRR	Co
Severe bradycardia	05-02 09:00	05-02 09:00	1	5	62	44	19	24	
Severe bradycardia	05-03 23:00	05-03 23:00	1	5	51	45	21	25	
Severe bradycardia	05-04 16:00	05-04 17:00	2	5	55	44	20	24	
Tachypnea	05-12 00:00	05-12 00:00	1	2	63	57	24	28	
Bradycardia	06-03 12:00	06-03 12:00	1	3	48	46	21	25	
Tachypnea	06-21 17:00	06-21 17:00	1	2	58	51	24	44	
Bradycardia	06-23 00:00	06-23 00:00	1	3	48	40	23	38	
Severe bradycardia	06-23 00:00	06-23 01:00	2	5	49	40	21	38	
Severe bradycardia	06-23 05:00	06-23 05:00	1	5	60	45	20	35	
Tachypnea	06-23 17:00	06-23 17:00	1	2	52	47	29	40	
Severe bradycardia	06-23 21:00	06-24 05:00	9	11	47	40	26	40	Y
Bradycardia	06-23 22:00	06-24 05:00	8	8	46	40	26	40	Y
Tachypnea	06-24 00:00	06-24 09:00	10	8	47	40	28	40	Y
Bradycardia	06-24 07:00	06-24 11:00	5	7	48	42	27	38	Y
Severe bradycardia	06-24 07:00	06-24 10:00	4	9	48	42	27	38	Y
Tachypnea	06-24 11:00	06-24 11:00	1	5	48	46	26	35	Y
Severe bradycardia	06-25 01:00	06-25 05:00	5	6	45	40	14	29	
Bradycardia	06-25 02:00	06-25 05:00	4	4	44	40	14	29	
Tachycardia	06-27 20:00	06-28 02:00	7	5	102	91	12	19	
Severe bradycardia	06-29 06:00	06-29 06:00	1	5	50	44	14	22	
Bradycardia	06-29 08:00	06-29 08:00	1	2	47	47	11	11	